```
No VM guests are running outdated hypervisor (qemu) binaries on
root@testPSQS01:/tmp# clickhouse-backup --version
Version:         2.6.10
Git Commit:      9975517043911be4a12ab97bee73d1219409b881
Build Date:      2025-04-01
root@testPSQS01:/tmp#
```

```
    debug: false
s3:
    access_key: "[redacted]"
    secret_key: "[redacted]"
    bucket: "test-barykin"
    endpoint: "https://storage.yandexcloud.net"
    region: us-east-1
```

```
CREATE DATABASE tst_bcp

Query id: 152e0379-3302-427e-8cdc-a9a173122d8e

Ok.
```

```
order by (product_id);

CREATE TABLE IF NOT EXISTS tst_bcp.product
(
    `product_id` UInt16,
    `product_name` String
)
ENGINE = MergeTree
ORDER BY product_id

Query id: c747d651-0bdc-489f-a044-a9d8f4285f34

Ok.

0 rows in set. Elapsed: 0.007 sec.

testPSQS01 :) INSERT INTO tst_bcp.product (product_id, product_name) VALUES
    (1, 'Laptop'),
    (2, 'Smartphone'),
    (3, 'Wireless Mouse'),
    (4, 'Mechanical Keyboard'),
    (5, 'USB-C Hub');

INSERT INTO tst_bcp.product (product_id, product_name) FORMAT Values

Query id: 130e1e51-e02b-4584-afab-4cce72fb2024

Ok.

5 rows in set. Elapsed: 0.004 sec.
```

```
CREATE TABLE IF NOT EXISTS tst_bcp.store
(
    `store_id` UInt16,
    `store_name` String
)
ENGINE = MergeTree
ORDER BY store_id

Query id: 83bac0ed-485a-4874-9b55-7fea8bcc1884

Ok.

0 rows in set. Elapsed: 0.009 sec.

testPSQS01 :) INSERT INTO tst_bcp.store (store_id, store_name) VALUES
    (1, 'Main Store'),
    (2, 'Downtown'),
    (3, 'Westside'),
    (4, 'Eastside');

INSERT INTO tst_bcp.store (store_id, store_name) FORMAT Values

Query id: b2b7ec09-5557-4aaf-bd31-b288bd579a14

Ok.

4 rows in set. Elapsed: 0.004 sec.
```

```
CREATE TABLE IF NOT EXISTS tst_bcp.sales
(
    `sale_id` String,
    `sale_date_str` LowCardinality(String),
    `store_id` UInt16,
    `product_id` UInt16,
    `quantity` UInt16,
    `sale_date` Date MATERIALIZED toDate(parseDateTimeBestEffort(sale_date_str)),
    `sale_yyyymm` UInt32 MATERIALIZED toYYYYMM(sale_date),
    `sale_dow` UInt8 MATERIALIZED toUInt8(toDayOfWeek(sale_date))
)
ENGINE = MergeTree
PARTITION BY sale_yyyymm
ORDER BY (sale_date, store_id, product_id)

Query id: c39b4416-6a67-48da-a478-41951387ae59

Ok.

0 rows in set. Elapsed: 0.009 sec.

testPSQS01 :) INSERT INTO tst_bcp.sales (sale_id, sale_date_str, store_id, product_id, quantity) VALUES
    ('SALE_0001', '15-01-2026', 1, 1, 2),
    ('SALE_0002', '15-01-2026', 2, 2, 1),
    ('SALE_0003', '20-01-2026', 3, 3, 3),
    ('SALE_0004', '05-02-2026', 1, 4, 1),
    ('SALE_0005', '12-02-2026', 4, 5, 4),
    ('SALE_0006', '18-02-2026', 2, 1, 2),
    ('SALE_0007', '01-03-2026', 3, 2, 1),
    ('SALE_0008', '10-03-2026', 4, 3, 2),
    ('SALE_0009', '22-03-2026', 1, 4, 3),
    ('SALE_0010', '28-03-2026', 2, 5, 1);

INSERT INTO tst_bcp.sales (sale_id, sale_date_str, store_id, product_id, quantity) FORMAT Values

Query id: 725e28a5-6a3c-4955-8ee6-8e5ede3a8684

Ok.

10 rows in set. Elapsed: 0.007 sec.
```

```
root@testPSQS01:/etc/clickhouse-backup# clickhouse-backup create --tables=tst_bcp.* first_bcp
2026-03-21 11:05:10.427 INF pkg/clickhouse/clickhouse.go:131 > clickhouse connection prepared: tcp://localhost:9000 run ping
2026-03-21 11:05:10.429 INF pkg/clickhouse/clickhouse.go:134 > clickhouse connection success: tcp://localhost:9000
2026-03-21 11:05:10.429 INF pkg/clickhouse/clickhouse.go:1123 > SELECT value FROM `system`.`build_options` where name='VERSION_INTEGER'
2026-03-21 11:05:10.431 INF pkg/clickhouse/clickhouse.go:1123 > SELECT data_path AS metadata_path FROM system.databases WHERE name = 'system' LIMIT 1
2026-03-21 11:05:10.433 INF pkg/clickhouse/clickhouse.go:1123 > SELECT name, engine FROM system.databases WHERE NOT match(name,'^(system|INFORMATION_SCHEMA|in
formation_schema|_temporary_and_external_tables)$') AND match(name,'^(tst_bcp)$')
2026-03-21 11:05:10.435 INF pkg/clickhouse/clickhouse.go:1123 > SHOW CREATE DATABASE `tst_bcp`
2026-03-21 11:05:10.436 INF pkg/clickhouse/clickhouse.go:1121 > SELECT name, count(*) as is_present FROM system.settings WHERE name IN (?, ?) GROUP BY name wi
th args []interface {}{"show_table_uuid_in_table_create_query_if_not_nil", "display_secrets_in_show_and_select"}
2026-03-21 11:05:10.441 INF pkg/clickhouse/clickhouse.go:1123 > SELECT name FROM system.databases WHERE engine IN ('MySQL','PostgreSQL','MaterializedPostgreSQ
L')
2026-03-21 11:05:10.442 INF pkg/clickhouse/clickhouse.go:1123 >    SELECT     countIf(name='data_path') is_data_path_present,     countIf(name='data_paths') i
s_data_paths_present,     countIf(name='uuid') is_uuid_present,     countIf(name='create_table_query') is_create_table_query_present,     countIf(name='total_
bytes') is_total_bytes_present    FROM system.columns WHERE database='system' AND table='tables'
2026-03-21 11:05:10.445 INF pkg/clickhouse/clickhouse.go:1123 > SELECT database, name, engine , data_paths , uuid , create_table_query , coalesce(total_bytes,
 0) AS total_bytes   FROM system.tables WHERE is_temporary = 0 AND match(concat(database,'.',name),'^tst_bcp\..*$')  ORDER BY total_bytes DESC SETTINGS show_t
able_uuid_in_table_create_query_if_not_nil=1
2026-03-21 11:05:10.476 INF pkg/clickhouse/clickhouse.go:1123 > SELECT data_path AS metadata_path FROM system.databases WHERE name = 'system' LIMIT 1
2026-03-21 11:05:10.478 INF pkg/clickhouse/clickhouse.go:1123 > SELECT count() as cnt FROM system.columns WHERE database='system' AND table='functions' AND na
me='create_query' SETTINGS empty_result_for_aggregation_by_empty_set=0
2026-03-21 11:05:10.481 INF pkg/clickhouse/clickhouse.go:1123 > SELECT name, create_query FROM system.functions WHERE create_query≠''
2026-03-21 11:05:10.492 INF pkg/clickhouse/clickhouse.go:1123 > SELECT countIf(name='type') AS is_disk_type_present, countIf(name='object_storage_type') AS is
_object_storage_type_present, countIf(name='free_space') AS is_free_space_present, countIf(name='disks') AS is_storage_policy_present FROM system.columns WHER
E database='system' AND table IN ('disks','storage_policies')
2026-03-21 11:05:10.495 INF pkg/clickhouse/clickhouse.go:1123 > SELECT d.path AS path, any(d.name) AS name, any(lower(if(d.type='ObjectStorage',d.object_stora
ge_type,d.type))) AS type, min(d.free_space) AS free_space, groupUniqArray(s.policy_name) AS storage_policies FROM system.disks AS d  LEFT JOIN (SELECT policy
_name, arrayJoin(disks) AS disk FROM system.storage_policies) AS s ON s.disk = d.name GROUP BY d.path
2026-03-21 11:05:10.500 INF pkg/clickhouse/clickhouse.go:1123 > SELECT name FROM system.user_directories WHERE type='replicated'
2026-03-21 11:05:10.502 INF pkg/clickhouse/clickhouse.go:1123 > SELECT JSONExtractString(params,'path') AS access_path FROM system.user_directories WHERE type
='local directory'
2026-03-21 11:05:10.504 INF pkg/clickhouse/clickhouse.go:1123 > SELECT data_path AS metadata_path FROM system.databases WHERE name = 'system' LIMIT 1
2026-03-21 11:05:10.510 INF pkg/backup/create.go:181 > done createBackupRBAC size=194B
2026-03-21 11:05:10.510 INF pkg/clickhouse/clickhouse.go:1121 > SELECT column, groupUniqArray(type) AS uniq_types FROM system.parts_columns WHERE active AND d
atabase=? AND table=? AND type NOT LIKE 'Enum%(%' AND type NOT LIKE 'Tuple(%' AND type NOT LIKE 'Nullable(Enum%(%' AND type NOT LIKE 'Nullable(Tuple(%' AND ty
```

```
    (UPDATE product_name = 'bad value' WHERE product_id = 3)

ALTER TABLE tst_bcp.product
    (UPDATE product_name = 'bad value' WHERE product_id = 3)

Query id: e465b989-b298-4400-aae3-c49cd941038e

Ok.

0 rows in set. Elapsed: 0.006 sec.

testPSQS01 :) ALTER TABLE tst_bcp.store DELETE WHERE store_id = 2;

ALTER TABLE tst_bcp.store
    (DELETE WHERE store_id = 2)

Query id: d8216387-4188-4e4e-9168-ba526f279c82

Ok.

0 rows in set. Elapsed: 0.006 sec.

testPSQS01 :) drop table tst_bcp.sales;

DROP TABLE tst_bcp.sales

Query id: b34f6be6-cc16-4028-bba4-3ee54f806755

Ok.

0 rows in set. Elapsed: 0.001 sec.
```

```
2026-03-21 11:12:12.405 INF pkg/clickhouse/clickhouse.go:336 > clickhouse connection closed
root@testPSQS01:/etc/clickhouse-backup# clickhouse-backup upload first_bcp
2026-03-21 11:12:38.379 INF pkg/clickhouse/clickhouse.go:131 > clickhouse connection prepared: tcp://localhost:9000 run ping
2026-03-21 11:12:38.381 INF pkg/clickhouse/clickhouse.go:134 > clickhouse connection success: tcp://localhost:9000
2026-03-21 11:12:38.381 INF pkg/clickhouse/clickhouse.go:1123 > SELECT value FROM `system`.`build_options` where name='VERSION_INTEGER'
2026-03-21 11:12:38.383 INF pkg/clickhouse/clickhouse.go:1123 > SELECT countIf(name='type') AS is_disk_type_present, countIf(name='object_storage_type') AS is
_object_storage_type_present, countIf(name='free_space') AS is_free_space_present, countIf(name='disks') AS is_storage_policy_present FROM system.columns WHER
E database='system' AND table IN ('disks','storage_policies')
2026-03-21 11:12:38.387 INF pkg/clickhouse/clickhouse.go:1123 > SELECT d.path AS path, any(d.name) AS name, any(lower(if(d.type='ObjectStorage',d.object_stora
ge_type,d.type))) AS type, min(d.free_space) AS free_space, groupUniqArray(s.policy_name) AS storage_policies FROM system.disks AS d  LEFT JOIN (SELECT policy
_name, arrayJoin(disks) AS disk FROM system.storage_policies) AS s ON s.disk = d.name GROUP BY d.path
2026-03-21 11:12:38.391 INF pkg/clickhouse/clickhouse.go:1123 > SELECT max(toInt64(bytes_on_disk * 1.02)) AS max_file_size FROM system.parts WHERE active SETT
INGS empty_result_for_aggregation_by_empty_set=0
2026-03-21 11:12:38.394 INF pkg/clickhouse/clickhouse.go:1123 > SELECT count() AS is_macros_exists FROM system.tables WHERE database='system' AND name='macros
'  SETTINGS empty_result_for_aggregation_by_empty_set=0
2026-03-21 11:12:38.396 INF pkg/clickhouse/clickhouse.go:1123 > SELECT macro, substitution FROM system.macros
2026-03-21 11:12:38.398 INF pkg/clickhouse/clickhouse.go:1123 > SELECT count() AS is_macros_exists FROM system.tables WHERE database='system' AND name='macros
'  SETTINGS empty_result_for_aggregation_by_empty_set=0
2026-03-21 11:12:38.400 INF pkg/clickhouse/clickhouse.go:1123 > SELECT macro, substitution FROM system.macros
2026-03-21 11:12:38.454 INF pkg/storage/general.go:163 > list_duration=19.012889
2026-03-21 11:12:38.564 INF pkg/backup/upload.go:184 > done data_size=10.00KiB duration=91ms metadata_size=460B operation=upload_table progress=1/3 table=tst_
bcp.product version=2.6.10
2026-03-21 11:12:38.749 INF pkg/backup/upload.go:184 > done data_size=37.50KiB duration=276ms metadata_size=919B operation=upload_table progress=2/3 table=tst
_bcp.sales version=2.6.10
2026-03-21 11:12:38.831 INF pkg/backup/upload.go:184 > done data_size=10.00KiB duration=267ms metadata_size=450B operation=upload_table progress=3/3 table=tst
_bcp.store version=2.6.10
2026-03-21 11:12:38.911 INF pkg/backup/upload.go:256 > done backup=first_bcp duration=533ms object_disk_size=0B operation=upload upload_size=62.98KiB version=
2.6.10
2026-03-21 11:12:38.911 INF pkg/clickhouse/clickhouse.go:336 > clickhouse connection closed
root@testPSQS01:/etc/clickhouse-backup#
```

```
drwxr-xr-x  3 clickhouse clickhouse 4096 Mar 21 11:12 first_bcp
root@testPSQS01:/etc/clickhouse-backup# clickhouse-backup restore first_bcp
2026-03-21 11:36:11.127 INF pkg/clickhouse/clickhouse.go:131 > clickhouse connection prepared: tcp://localhost:9000 run ping
2026-03-21 11:36:11.128 INF pkg/clickhouse/clickhouse.go:134 > clickhouse connection success: tcp://localhost:9000
2026-03-21 11:36:11.128 INF pkg/clickhouse/clickhouse.go:1123 > SELECT value FROM `system`.`build_options` where name='VERSION_INTEGER'
2026-03-21 11:36:11.131 INF pkg/clickhouse/clickhouse.go:1123 > SELECT countIf(name='type') AS is_disk_type_present, countIf(name='object_storage_type') AS is
_object_storage_type_present, countIf(name='free_space') AS is_free_space_present, countIf(name='disks') AS is_storage_policy_present FROM system.columns WHER
E database='system' AND table IN ('disks','storage_policies')
2026-03-21 11:36:11.134 INF pkg/clickhouse/clickhouse.go:1123 > SELECT d.path AS path, any(d.name) AS name, any(lower(if(d.type='ObjectStorage',d.object_stora
ge_type,d.type))) AS type, min(d.free_space) AS free_space, groupUniqArray(s.policy_name) AS storage_policies FROM system.disks AS d  LEFT JOIN (SELECT policy
_name, arrayJoin(disks) AS disk FROM system.storage_policies) AS s ON s.disk = d.name GROUP BY d.path
2026-03-21 11:36:11.138 INF pkg/clickhouse/clickhouse.go:1121 > CREATE DATABASE IF NOT EXISTS `tst_bcp` ENGINE = Atomic with args []interface {}{[]interface {
}(nil)}
2026-03-21 11:36:11.140 INF pkg/clickhouse/clickhouse.go:1123 > SELECT engine FROM system.databases WHERE name = 'tst_bcp'
2026-03-21 11:36:11.144 INF pkg/clickhouse/clickhouse.go:1123 > DROP TABLE IF EXISTS `tst_bcp`.`product` NO DELAY
2026-03-21 11:36:11.146 INF pkg/clickhouse/clickhouse.go:1123 > SELECT engine FROM system.databases WHERE name = 'tst_bcp'
2026-03-21 11:36:11.147 INF pkg/clickhouse/clickhouse.go:1123 > DROP TABLE IF EXISTS `tst_bcp`.`sales` NO DELAY
2026-03-21 11:36:11.148 INF pkg/clickhouse/clickhouse.go:1123 > SELECT engine FROM system.databases WHERE name = 'tst_bcp'
2026-03-21 11:36:11.149 INF pkg/clickhouse/clickhouse.go:1123 > DROP TABLE IF EXISTS `tst_bcp`.`store` NO DELAY
2026-03-21 11:36:11.150 INF pkg/clickhouse/clickhouse.go:1123 > CREATE DATABASE IF NOT EXISTS `tst_bcp`
2026-03-21 11:36:11.151 INF pkg/clickhouse/clickhouse.go:1123 > CREATE TABLE tst_bcp.product UUID '6b5ed012-1021-492c-a46f-69e30852608f' (`product_id` UInt16,
 `product_name` String) ENGINE = MergeTree ORDER BY product_id SETTINGS index_granularity = 8192
2026-03-21 11:36:11.161 INF pkg/clickhouse/clickhouse.go:1123 > CREATE TABLE tst_bcp.sales UUID '16fe12d7-a01a-4787-8d86-835813686ad5' (`sale_id` String, `sal
e_date_str` LowCardinality(String), `store_id` UInt16, `product_id` UInt16, `quantity` UInt16, `sale_date` Date MATERIALIZED toDate(parseDateTimeBestEffort(sa
le_date_str)), `sale_yyyymm` UInt32 MATERIALIZED toYYYYMM(sale_date), `sale_dow` UInt8 MATERIALIZED toUInt8(toDayOfWeek(sale_date))) ENGINE = MergeTree PARTIT
ION BY sale_yyyymm ORDER BY (sale_date, store_id, product_id) SETTINGS index_granularity = 8192
2026-03-21 11:36:11.168 INF pkg/clickhouse/clickhouse.go:1123 > CREATE TABLE tst_bcp.store UUID 'ef9c8c86-2e4a-4819-9a0e-d6753fac9824' (`store_id` UInt16, `st
ore_name` String) ENGINE = MergeTree ORDER BY store_id SETTINGS index_granularity = 8192
2026-03-21 11:36:11.179 INF pkg/backup/restore.go:945 > done backup=first_bcp duration=39ms operation=restore_schema
```

```
SHOW TABLES

Query id: 9afeaa19-4933-40eb-a67c-508dc9e05dbf

    ┌─name────┐
1.  │ product │
2.  │ sales   │
3.  │ store   │
    └─────────┘

3 rows in set. Elapsed: 0.002 sec.

testPSQS01 :) select count(*) from store;

SELECT count(*)
FROM store

Query id: 100f2459-ee09-43e0-8836-4e1cc110e122

    ┌─count()─┐
1.  │       4 │
    └─────────┘

1 row in set. Elapsed: 0.002 sec.

testPSQS01 :) select * from product where product_id = 3;

SELECT *
FROM product
WHERE product_id = 3

Query id: 311deaf0-296e-4771-a5cc-f38b2b4ff306

    ┌─product_id─┬─product_name──┐
1.  │          3 │ Wireless Mouse │
    └────────────┴───────────────┘
```